

A Course End Project Report on

Pharmacy Management System

Submitted by

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Abstract

Traditional pharmacy management involves several activities, including medicine inventory maintenance, sales recording, stock availability monitoring, billing, and purchase record management. Handling these activities manually can be time-consuming and may lead to errors, data duplication, and difficulties in tracking stock and retrieving information. The Pharmacy Management System addresses these challenges by providing a computerized platform to manage pharmacy operations efficiently.

The proposed system maintains medicine details, manages inventory, records sales transactions, generates bills, and retrieves information whenever required. Medicine records include details such as medicine name, identification number, category, price, available quantity, and expiry date. The inventory module helps staff monitor stock levels, while the billing module calculates total costs based on selected medicines, quantities, and prices.

The system aims to improve the efficiency, accuracy, and reliability of pharmacy operations while reducing manual record-keeping. By maintaining structured digital records, it simplifies information retrieval and supports better inventory management. Future enhancements may include online medicine ordering, supplier management, automated expiry notifications, sales reports, secure user authentication, and role-based access control. Overall, the system provides a convenient and cost-effective approach to pharmacy management.

Keywords: Pharmacy Management System, Medicine Inventory, Stock Management, Billing, Sales Records, Database Management, Expiry Date Monitoring, Digital Record-Keeping.

Chapter 1

Introduction

A pharmacy handles various activities, including medicine purchasing, inventory maintenance, billing, sales recording, and maintaining medicine-related information. Managing these activities manually can be time-consuming and may lead to errors, duplication of data, and difficulties in tracking stock quantities and retrieving information. Therefore, a computerized system is useful for improving the efficiency of pharmacy operations.

The Pharmacy Management System is designed to computerize important pharmacy activities and provide an organized method for managing medicine information. It maintains details such as medicine name, identification number, category, price, available quantity, and expiry date. These records help pharmacy staff access medicine information quickly and maintain updated inventory details.

Inventory management is an important feature of the system. It helps monitor medicine quantities and update stock whenever medicines are purchased or sold. This makes it easier to identify medicines that require replenishment and check their availability. Recording expiry dates can also help identify medicines approaching their expiry dates when this functionality is implemented.

The system also supports billing and sales management by calculating the total cost based on medicine prices and quantities, generating bills, and maintaining transaction records. A centralized database helps organize medicine details and sales information, making record retrieval and maintenance more convenient.

The main objective of this project is to reduce manual effort, improve record accuracy, and simplify routine pharmacy operations. It demonstrates the application of software development and database management concepts to solve practical problems. Future enhancements may include online medicine ordering, supplier management, automated expiry notifications, sales reports, and secure user access. Overall, the system aims to provide an efficient, reliable, and organized solution for pharmacy management.

Chapter 2

Literature Review

Traditional pharmacy management methods generally depend on paper registers and manual calculations to maintain medicine details, inventory records, purchase information, and sales transactions. These methods require considerable time to update records and prepare bills. They may also lead to calculation errors, duplication of information, and difficulties in identifying available medicines and monitoring stock levels.

Database-driven management systems provide a systematic approach to storing, organizing, updating, and retrieving information. By maintaining records in a structured format, these systems improve data consistency and make information retrieval easier. They also support efficient management of medicine details, including prices, quantities, categories, and expiry dates.

Inventory management applications play an important role in monitoring stock availability and identifying products that require replenishment. Maintaining updated inventory records helps staff manage medicine quantities and plan purchases. Similarly, computerized billing applications automate price calculations, generate bills, and maintain transaction records. These functionalities reduce repetitive manual work and improve the efficiency of routine business operations.

The Pharmacy Management System combines medicine management, inventory tracking, and billing functionalities into a single application. It aims to provide an organized platform for maintaining pharmacy records and simplifying daily activities. The system can also be extended with features such as expiry notifications, supplier management, and sales reporting to improve its usefulness.

Although computerized systems offer several advantages, their effectiveness depends on accurate data entry, appropriate database design, reliable software implementation, and suitable access controls. Therefore, the proposed system focuses on organized record maintenance, efficient information retrieval, and simplified pharmacy operations.

Chapter 3

Objectives

The main objective of the Pharmacy Management System is to simplify and improve pharmacy operations by maintaining medicine information, managing inventory, recording sales, and generating bills through a computerized application. The specific objectives of the project are:

- **Centralized Medicine Database:** To maintain medicine details, including medicine name, identification number, category, price, quantity, and expiry date, in a centralized database.
- **Inventory Management:** To manage medicine stock and availability by maintaining updated quantity records whenever medicines are purchased or sold.
- **Billing and Sales Management:** To record medicine sales, calculate the total bill based on medicine prices and quantities, and generate bills accurately.
- **Purchase and Sales Records:** To maintain organized purchase and sales records for convenient reference and transaction tracking.
- **Medicine Search and Retrieval:** To provide a convenient way to search for medicines and retrieve relevant information whenever required.
- **Stock and Expiry Monitoring:** To help identify medicines with low stock levels and monitor expiry dates, where these features are implemented.
- **Reduction of Manual Work:** To reduce paperwork, repetitive data entry, and manual calculations involved in routine pharmacy activities.
- **Improved Efficiency and Accuracy:** To improve the organization, accessibility, and accuracy of pharmacy records and support efficient daily operations.

The overall aim is to provide a simple, reliable, and organized system that helps pharmacy staff manage medicine information, inventory, and sales more efficiently.

Chapter 4

Methodology

The Pharmacy Management System follows a systematic approach to recording, processing, storing, and retrieving pharmacy information. The application accepts user inputs, validates the entered data, processes medicine and sales information, and stores or retrieves records from the database. The methodology consists of system design, medicine management, inventory management, billing, database operations, and report generation.

4.1 Block Diagram

The system consists of the following major components:

- **User Interface:** Allows authorized users to interact with the application and enter or retrieve data.
- **Application Logic:** Processes medicine details, stock updates, sales transactions, and billing operations.
- **Database:** Stores medicine details, inventory information, and transaction records.
- **Reporting Module:** Displays or generates reports using the information maintained by the system.

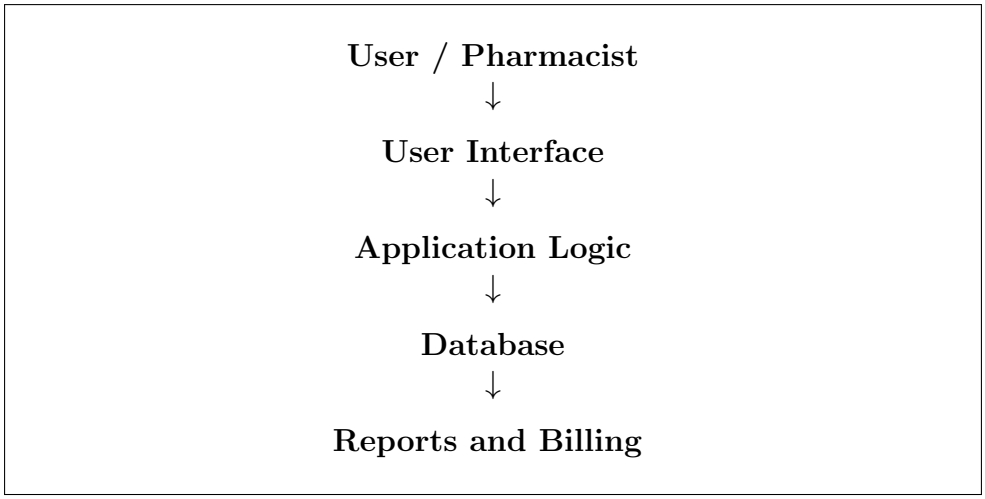


Figure 4.1: Block Diagram of Pharmacy Management System

4.2 System Design and Components

The Pharmacy Management System is divided into functional modules to simplify the management of pharmacy activities.

4.2.1 Medicine Management

This module maintains medicine information, including medicine name, identification number, category, price, available quantity, and expiry date. Authorized users can add new medicine records, search for existing medicines, and update stored information. Maintaining organized medicine records makes information retrieval easier.

4.2.2 Inventory Management

The inventory module tracks the quantity of medicines available in stock. Stock quantities are updated when purchases or sales are recorded. Monitoring inventory helps pharmacy staff identify medicines that require replenishment. Expiry-date monitoring can also be included to identify medicines approaching their expiry dates, depending on the implemented features.

4.2.3 Billing and Sales Management

The billing module records medicine sales and calculates the total bill based on the selected medicines, quantities, and stored prices. The system can maintain transaction records for future reference. Updating the stock after a sale helps keep inventory information consistent with recorded transactions.

4.2.4 Database Management

The database stores medicine details, inventory information, and sales or purchase records. A structured database organizes the information and supports efficient insertion, updating, searching, and retrieval of records. Input validation and appropriate database operations help maintain data consistency.

4.2.5 Reporting Module

The reporting module presents information obtained from stored records. Depending on the implemented functionality, reports may include medicine availability, low-stock items, sales transactions, and purchase details. These reports help users review pharmacy activities and access relevant information.

4.3 Working Principle

The system operates through the following steps:

1. The authorized user opens the Pharmacy Management System.
2. The user enters medicine details or searches for existing medicine records.
3. The application validates the input and retrieves the relevant information from the database.
4. For a sale, the user selects the required medicines and enters their quantities.
5. The system calculates the total bill using the stored medicine prices and selected quantities.
6. The sales transaction is recorded, and the corresponding stock quantity is updated.
7. The user can retrieve stored information and access available reports.

4.4 Software Implementation

The software technologies used depend on the actual project implementation. The relevant technologies should be specified according to the application developed.

- **Programming Language:** Python, Java, PHP, or another language used in the project.

- **User Interface:** Web-based or desktop interface.
- **Database:** MySQL, SQLite, or another database system used to store application records.
- **Development Tools:** The IDE, frameworks, and supporting tools used during development.

The application connects the user interface to the database through the application logic. Input validation should be applied to identify incomplete medicine records, invalid quantities, and incorrect prices. The implemented system should also handle database errors appropriately to improve reliability and maintain consistent records.

Chapter 5

Source Code

5.1 Pharmacy Management System in Java

The following source code implements medicine management, stock updates, billing, and sales records.

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class PharmacyManagementSystem {

    static Scanner sc = new Scanner(System.in);
    static HashMap<Integer, Medicine> medicines = new HashMap<>();
    static ArrayList<String> salesRecords = new ArrayList<>();

    static class Medicine {
        int id;
        String name;
        String category;
        double price;
        int quantity;
        String expiryDate;

        Medicine(int id, String name, String category,
                 double price, int quantity, String expiryDate) {
            this.id = id;
            this.name = name;
            this.category = category;
            this.price = price;
            this.quantity = quantity;
            this.expiryDate = expiryDate;
        }

```

```
30          void display() {
31              System.out.printf(
32                  "%-8d %-20s %-15s Rs.%-10.2f %-10d %-12s%n",
33                  id, name, category, price, quantity, expiryDate
34              );
35          }
36      }
37
38      public static void main(String[] args) {
39          medicines.put(101, new Medicine(
40              101, "Paracetamol", "Tablet", 20.00, 100, "12/2027"));
41          medicines.put(102, new Medicine(
42              102, "Amoxicillin", "Capsule", 60.00, 50, "08/2027"));
43          medicines.put(103, new Medicine(
44              103, "Cetirizine", "Tablet", 15.00, 30, "10/2027"));
45
46          int choice;
47
48          do {
49              System.out.println("\\n==================================")
                    ;
50              System.out.println(" PHARMACY MANAGEMENT SYSTEM");
51              System.out.println("==================================");
52              System.out.println("1. Add Medicine");
53              System.out.println("2. View All Medicines");
54              System.out.println("3. Search Medicine");
55              System.out.println("4. Update Medicine Stock");
56              System.out.println("5. Generate Bill");
57              System.out.println("6. View Low-Stock Medicines");
58              System.out.println("7. View Sales Records");
59              System.out.println("8. Exit");
60              System.out.print("Enter your choice: ");
61
62              choice = readInt();
63
64              switch (choice) {
65                  case 1: addMedicine(); break;
66                  case 2: viewMedicines(); break;
67                  case 3: searchMedicine(); break;
68                  case 4: updateStock(); break;
69                  case 5: generateBill(); break;
70                  case 6: viewLowStock(); break;
71                  case 7: viewSalesRecords(); break;
72                  case 8:
73                      System.out.println("Thank you for using the system!
                            ");
74                      break;
75                  default:
76                      System.out.println("Invalid choice. Try again.");
77              }
```

```
78          } while (choice != 8);
79
80          sc.close();
81      }
82
83      static int readInt() {
84          while (!sc.hasNextInt()) {
85              System.out.print("Enter a valid whole number: ");
86              sc.next();
87          }
88          int value = sc.nextInt();
89          sc.nextLine();
90          return value;
91      }
92
93      static double readDouble() {
94          while (!sc.hasNextDouble()) {
95              System.out.print("Enter a valid number: ");
96              sc.next();
97          }
98          double value = sc.nextDouble();
99          sc.nextLine();
100         return value;
101     }
102
103     static void addMedicine() {
104         System.out.print("Enter medicine ID: ");
105         int id = readInt();
106
107         if (medicines.containsKey(id)) {
108             System.out.println("Medicine ID already exists.");
109             return;
110         }
111
112         System.out.print("Enter medicine name: ");
113         String name = sc.nextLine().trim();
114
115         System.out.print("Enter category: ");
116         String category = sc.nextLine().trim();
117
118         System.out.print("Enter price: ");
119         double price = readDouble();
120
121         System.out.print("Enter quantity: ");
122         int quantity = readInt();
123
124         System.out.print("Enter expiry date (MM/YYYY): ");
125         String expiry = sc.nextLine().trim();
126
127         if (name.isEmpty() || category.isEmpty() || expiry.isEmpty()
```

```
                || price < 0 || quantity < 0) {
            System.out.println("Invalid details. Medicine not added.");
            return;
        }

        medicines.put(id, new Medicine(
            id, name, category, price, quantity, expiry));
        System.out.println("Medicine added successfully.");
    }

    static void viewMedicines() {
        if (medicines.isEmpty()) {
            System.out.println("No medicines available.");
            return;
        }
        printHeader();
        for (Medicine m : medicines.values()) {
            m.display();
        }
    }

    static void printHeader() {
        System.out.printf("%-8s %-20s %-15s %-13s %-10s %-12s%n",
                "ID", "Name", "Category", "Price", "Quantity", "Expiry"
                    );
        System.out.println(
                "
                    ----------------------------------------------------------
                    ");
    }

    static void searchMedicine() {
        System.out.print("Enter medicine ID: ");
        int id = readInt();
        Medicine m = medicines.get(id);

        if (m != null) {
            printHeader();
            m.display();
        } else {
            System.out.println("Medicine not found.");
        }
    }

    static void updateStock() {
        System.out.print("Enter medicine ID: ");
        int id = readInt();
        Medicine m = medicines.get(id);

        if (m == null) {
```

```
175             System.out.println("Medicine not found.");
176             return;
177         }
178 
179         System.out.println("Current stock: " + m.quantity);
180         System.out.println("1. Add stock");
181         System.out.println("2. Set stock quantity");
182         System.out.print("Choose an option: ");
183         int option = readInt();
184 
185         if (option == 1) {
186             System.out.print("Enter quantity to add: ");
187             int amount = readInt();
188             if (amount <= 0) {
189                 System.out.println("Quantity must be positive.");
190                 return;
191             }
192             m.quantity += amount;
193             System.out.println("Stock increased successfully.");
194         } else if (option == 2) {
195             System.out.print("Enter new stock quantity: ");
196             int amount = readInt();
197             if (amount < 0) {
198                 System.out.println("Quantity cannot be negative.");
199                 return;
200             }
201             m.quantity = amount;
202             System.out.println("Stock updated successfully.");
203         } else {
204             System.out.println("Invalid option.");
205             return;
206         }
207         System.out.println("Updated stock: " + m.quantity);
208     }
209 
210     static void generateBill() {
211         HashMap<Integer, Integer> cart = new HashMap<>();
212         System.out.println("\\nEnter medicine IDs and quantities.");
213         System.out.println("Enter ID 0 to finish.");
214 
215         while (true) {
216             System.out.print("Medicine ID: ");
217             int id = readInt();
218             if (id == 0) break;
219 
220             Medicine m = medicines.get(id);
221             if (m == null) {
222                 System.out.println("Medicine not found.");
223                 continue;
224             }
```

```
225
226             System.out.println("Medicine: " + m.name);
227             System.out.println("Available stock: " + m.quantity);
228             System.out.print("Enter quantity: ");
229             int quantity = readInt();
230
231             if (quantity <= 0) {
232                 System.out.println("Quantity must be positive.");
233                 continue;
234             }
235
236             int requested = cart.getOrDefault(id, 0) + quantity;
237             if (requested > m.quantity) {
238                 System.out.println("Insufficient stock. Item not added.
                    ");
239                 continue;
240             }
241
242             cart.put(id, requested);
243             System.out.println("Item added to bill.");
244         }
245
246         if (cart.isEmpty()) {
247             System.out.println("No items selected. Bill cancelled.");
248             return;
249         }
250
251         double total = 0;
252         System.out.println("\\n========== PHARMACY BILL ==========");
253         System.out.printf("%-20s %-8s %-10s %-10s%n",
254                 "Medicine", "Qty", "Price", "Amount");
255         System.out.println("-----------------------------------");
256
257         for (Map.Entry<Integer, Integer> entry : cart.entrySet()) {
258             Medicine m = medicines.get(entry.getKey());
259             int quantity = entry.getValue();
260             double amount = m.price * quantity;
261             System.out.printf("%-20s %-8d %-10.2f %-10.2f%n",
262                     m.name, quantity, m.price, amount);
263             total += amount;
264         }
265
266         for (Map.Entry<Integer, Integer> entry : cart.entrySet()) {
267             Medicine m = medicines.get(entry.getKey());
268             m.quantity -= entry.getValue();
269         }
270
271         System.out.println("-----------------------------------");
272         System.out.printf("Total Amount: Rs. %.2f%n", total);
273         System.out.println("Bill generated successfully.");
```

```
274             System.out.println("==================================");
275
276             salesRecords.add("Total: Rs. " + String.format("%.2f", total)
277                     + " | Items: " + cart.size());
278         }
279
280         static void viewLowStock() {
281             boolean found = false;
282             System.out.println("\\nMedicines with stock below 10:");
283             printHeader();
284
285             for (Medicine m : medicines.values()) {
286                 if (m.quantity < 10) {
287                     m.display();
288                     found = true;
289                 }
290             }
291             if (!found) {
292                 System.out.println("No low-stock medicines found.");
293             }
294         }
295
296         static void viewSalesRecords() {
297             System.out.println("\\n========== SALES RECORDS ==========");
298             if (salesRecords.isEmpty()) {
299                 System.out.println("No sales recorded.");
300                 return;
301             }
302             for (int i = 0; i < salesRecords.size(); i++) {
303                 System.out.println((i + 1) + ". " + salesRecords.get(i));
304             }
305         }
306 }
```

Listing 5.1: Java Source Code for Pharmacy Management System

Chapter 6

Results and Output

6.1 Main Menu

When the program is executed, the following menu is displayed.

```
==================================
PHARMACY MANAGEMENT SYSTEM
==================================
1.  Add Medicine
2.  View All Medicines
3.  Search Medicine
4.  Update Medicine Stock
5.  Generate Bill
6.  View Low-Stock Medicines
7.  View Sales Records
8.  Exit
Enter your choice:
```

Figure 6.1: Main Menu of Pharmacy Management System

6.2 View All Medicines

When the user selects option 2, the available medicine records are displayed.

```
ID    Name            Category    Price     Quantity    Expiry
-------------------------------------------------------------------------
101    Paracetamol    Tablet    Rs.20.00    100    12/2027
102    Amoxicillin    Capsule    Rs.60.00    50    08/2027
103    Cetirizine    Tablet    Rs.15.00    30    10/2027
```

Figure 6.2: Medicine Inventory Output

6.3 Generate Bill

The billing module calculates the total amount based on the selected medicines, their prices, and quantities.

```
========== PHARMACY BILL ==========
Medicine     Qty    Price    Amount
-----------------------------------
Paracetamol    2    20.00    40.00
Amoxicillin    1    60.00    60.00
-----------------------------------
Total Amount:  Rs.  100.00
Bill generated successfully.
===================================
```

Figure 6.3: Sample Billing Output

6.4 Low-Stock Medicines

When the user selects option 6, the system displays medicines whose stock quantity is below 10.

```
Medicines with stock below 10:
No low-stock medicines found.
```

Figure 6.4: Low-Stock Medicines Output

6.5 Discussion

The sample outputs demonstrate the main functionalities of the Pharmacy Management System, including medicine listing, billing, and low-stock checking. After a successful sale, the corresponding medicine quantities are reduced automatically. The displayed results are illustrative examples of program execution; actual output depends on the operations performed by the user.

Chapter 7

Mapping POs and SDGs

7.1 Program Outcomes (POs) Mapping

The Pharmacy Management System applies software development and database management concepts to simplify pharmacy operations. The project contributes to selected Program Outcomes by addressing practical problems through a computerized solution.

Table 7.1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	**Program Outcome**	**Relevance**
PO1	Engineering Knowledge	Applies programming, database management, and software engineering concepts to develop the system.
PO2	Problem Analysis	Identifies challenges in manual pharmacy operations, including record maintenance, stock tracking, and billing.
PO3	Design/Development of Solutions	Designs a computerized solution for managing medicine information, inventory, and sales transactions.
PO5	Modern Tool Usage	Uses appropriate programming languages, development environments, and database technologies to implement the application.
PO6	Engineer and Society	Supports organized pharmacy operations and improves access to medicine and transaction information.

7.2 Sustainable Development Goals (SDGs) Mapping

The Pharmacy Management System can contribute to selected Sustainable Development Goals by supporting organized medicine management, digital innovation, and improved inventory monitoring.

Table 7.2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	**Goal**	**Relevance**
SDG 3	Good Health and Well-Being	Supports organized medicine inventory and pharmacy operations, helping staff manage medicine information and availability more efficiently.
SDG 9	Industry, Innovation and Infrastructure	Promotes the use of digital technology and software solutions to improve business management processes.
SDG 12	Responsible Consumption and Production	Supports stock monitoring and may help reduce avoidable medicine waste through improved inventory and expiry date tracking, where implemented.

The mapping demonstrates the potential contribution of the Pharmacy Management System to software engineering outcomes and selected Sustainable Development Goals. The extent of these contributions depends on the features implemented and the system's actual usage.

Chapter 8

Conclusion

The Pharmacy Management System provides a computerized approach to managing medicine information, inventory, billing, and sales records. It aims to simplify routine pharmacy activities by maintaining organized digital records and reducing dependence on manual record-keeping. The system is intended to make information retrieval more convenient and improve the efficiency of pharmacy operations.

The project demonstrates how programming and database management technologies can be applied to address practical problems in pharmacy management. The medicine management module helps organize medicine details, while the inventory module supports stock monitoring and the billing module facilitates sales calculations and transaction recording. Together, these functions provide a structured approach to managing essential pharmacy activities.

The effectiveness and reliability of the system depend on the quality of its implementation, database design, input validation, and testing. Testing medicine records, stock updates, bill calculations, and transaction handling is necessary to verify that the system operates as expected. Accurate data entry and proper maintenance are also important for keeping the stored information reliable.

Future enhancements may include supplier integration, automated expiry notifications, low-stock alerts, secure user roles, online medicine ordering, database backup facilities, and improved reporting features. These additions could further improve the usefulness, security, and scalability of the application.

In conclusion, the Pharmacy Management System aims to provide a simple, organized, and efficient solution for managing pharmacy operations. With appropriate implementation and testing, it can help reduce repetitive manual work, improve record management, and support the convenient handling of medicine inventory and sales information.

Bibliography

[1] R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th ed., Pearson, 2016.

[2] I. Sommerville, *Software Engineering*, 10th ed., Pearson, 2015.

[3] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed., McGraw-Hill, 2019.

[4] A. Silberschatz, H. F. Korth, and S. Sudarshan, *Database System Concepts*, 7th ed., McGraw-Hill, 2019.

[5] T. Connolly and C. Begg, *Database Systems: A Practical Approach to Design, Implementation, and Management*, 6th ed., Pearson, 2014.

[6] R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd ed., McGraw-Hill, 2003.

[7] K. C. Laudon and J. P. Laudon, *Management Information Systems: Managing the Digital Firm*, Pearson.

[8] A. S. Tanenbaum and M. van Steen, *Distributed Systems: Principles and Paradigms*, 2nd ed., Prentice Hall, 2007.

[9] M. Fowler, *Patterns of Enterprise Application Architecture*, Addison-Wesley, 2002.

[10] R. C. Martin, *Clean Code: A Handbook of Agile Software Craftsmanship*, Prentice Hall, 2008.